

Analysis III - CheatSheet

Special Version - BA3 2.0

Notes by Ali EL AZDI

January 12rd, 2026

Regular Curve. A continuously differentiable map $\gamma : [a, b] \rightarrow \mathbb{R}^n$ is *regular* if $\gamma'(t) \neq 0$ for all $t \in [a, b]$.

Simple Curve. A continuous map $\gamma : [a, b] \rightarrow \mathbb{R}^n$ is *simple* if it does not intersect itself (except possibly at endpoints): $\gamma(t_1) = \gamma(t_2) \implies t_1 = t_2$

Simply Connected Domain. An open set $\Omega \subseteq \mathbb{R}^n$ is *simply connected* if for any two continuous curves $\gamma_0, \gamma_1 : [a, b] \rightarrow \Omega$ with $\gamma_0(a) = \gamma_1(a)$ and $\gamma_0(b) = \gamma_1(b)$, there exists a continuous homotopy $\Gamma : [a, b] \times [0, 1] \rightarrow \Omega$ that deforms γ_0 into γ_1 within Ω while keeping the endpoints fixed.

Curve in $\mathbb{R}^n$: Given $\gamma : [a, b] \rightarrow \mathbb{R}^n$, $\int_\gamma F \cdot d\mathbf{l} = \int_a^b \langle F(\gamma(t)), \gamma'(t) \rangle dt$, $\int_\gamma f(\mathbf{x}) ds = \int_a^b f(\gamma(t)) \|\gamma'(t)\| dt$.

$$\begin{aligned} \nabla \cdot (\vec{A} \times \vec{B}) &= (\nabla \times \vec{A}) \cdot \vec{B} - \vec{A} \cdot (\nabla \times \vec{B}) \\ \operatorname{div}(\nabla f) &= \Delta f \\ \operatorname{rot}(\nabla f) &= 0 \\ \operatorname{div}(\operatorname{rot} \vec{F}) &= 0 \\ \nabla(fg) &= f \nabla g + g \nabla f \\ \operatorname{rot}(f \vec{F}) &= \nabla f \times \vec{F} + f \operatorname{rot}(\vec{F}) \\ \operatorname{div}(f \vec{F}) &= f \operatorname{div}(\vec{F}) + \langle \vec{F}, \nabla f \rangle \end{aligned}$$

Surface in $\mathbb{R}^3$: Given $\sigma : D \subset \mathbb{R}^2 \rightarrow \mathbb{R}^3$, $\sigma(u, v)$,

$$\iint_\sigma F \cdot d\mathbf{S} = \iint_D \langle F(\sigma(u, v)), \frac{\partial \sigma}{\partial u} \times \frac{\partial \sigma}{\partial v} \rangle du dv. \quad \iint_\sigma f(\mathbf{x}) ds = \iint_D f(\sigma(u, v)) \left\| \frac{\partial \sigma}{\partial u} \times \frac{\partial \sigma}{\partial v} \right\| du dv.$$

Conservative Fields and Path Independence

A map $F : \Omega \subseteq \mathbb{R}^n \rightarrow \mathbb{R}^n$ is *conservative* if there exists a differentiable function $\varphi : \Omega \rightarrow \mathbb{R}$ such that $\nabla \varphi = F$.

F is conservative on $\Omega \iff \forall \Gamma_1, \Gamma_2 \subseteq \Omega$ reg. curves from A to B , $\int_{\Gamma_1} F \cdot d\mathbf{l} = \int_{\Gamma_2} F \cdot d\mathbf{l}$, $\iff \forall \Gamma \subseteq \Omega$ reg. closed curve, $\int_\Gamma F \cdot d\mathbf{l} = 0$.

is F Conservative over Ω ?

1 - Compute curl F . **2 - is Ω Simply Connected ?**

- If $\operatorname{curl} F \neq 0$, F is **not** conservative.

- If $\operatorname{curl} F = 0$, proceed to Step 2.

$$\text{2D: } \nabla \times F = \begin{bmatrix} \frac{\partial F_y}{\partial x} - \frac{\partial F_x}{\partial y} \end{bmatrix}$$

$$\text{3D: } \nabla \times F = \begin{bmatrix} \frac{\partial F_z}{\partial y} - \frac{\partial F_y}{\partial z} \\ \frac{\partial F_x}{\partial z} - \frac{\partial F_z}{\partial x} \\ \frac{\partial F_y}{\partial x} - \frac{\partial F_x}{\partial y} \end{bmatrix}$$

3 - Circulation Method:

1. Select closed curves Γ around each hole in Ω .
2. Compute $\oint_\Gamma F \cdot d\mathbf{l}$.
3. If any integral $\neq 0$, F is **not** conservative.
4. If all integrals = 0, proceed to Step 4.

4 - Finding the Potential

1. $\varphi(x, y, z) = \int_{x_0}^x F_1(t, y, z) dt + \alpha(y, z)$
1. Determine $\alpha(y, z)$ such that $\nabla \varphi = F$.
2. If successful, φ is the potential function for F , confirming F is conservative.

$$\gamma(t) = (x(t), y(t)), \mathbf{n}(t) = \frac{(y'(t), -x'(t))}{\|\gamma'(t)\|} \quad \sigma(u, v) = (x(u, v), y(u, v), z(u, v)), \mathbf{n}(u, v) = \frac{\frac{\partial \sigma}{\partial u} \times \frac{\partial \sigma}{\partial v}}{\left\| \frac{\partial \sigma}{\partial u} \times \frac{\partial \sigma}{\partial v} \right\|}, \text{ ! not necessarily outward}$$

Vector Calculus Theorems

Green's Theorem (Plane) Let $\Omega \subseteq \mathbb{R}^2$ be a regular domain with a positively oriented boundary $\partial\Omega$, and $F \in C^1(\overline{\Omega}, \mathbb{R}^2)$. Then

$$\iint_\Omega \operatorname{curl}(F(x, y)) dx dy = \int_{\partial\Omega} F \cdot d\mathbf{l}$$

Divergence Theorem (Space) Let $\Omega \subseteq \mathbb{R}^3$ be a regular domain, $n : \partial\Omega \rightarrow \mathbb{R}^3$ a continuous outward unit normal vector field, and $F \in C^1(\overline{\Omega}, \mathbb{R}^3)$. Then

$$\iiint_\Omega \operatorname{div} F(x, y, z) dx dy dz = \iint_{\partial\Omega} F \cdot n ds = \iint_A \langle F(\sigma); \frac{\partial \sigma}{\partial u} \wedge \frac{\partial \sigma}{\partial v} \rangle du dv$$

Polar (2D)

$$|\det J| = r$$

$$\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} r \cos \theta \\ r \sin \theta \end{bmatrix};$$

$$r \geq 0, \theta \in [0, 2\pi].$$

Cylindrical (3D)

$$|\det J| = r$$

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} r \cos \theta \\ r \sin \theta \\ z \end{bmatrix};$$

$$r \geq 0, \theta \in [0, 2\pi], z \in (-\infty, \infty).$$

Spherical (3D)

$$|\det J| = \rho^2 \sin \phi$$

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} \rho \sin \phi \cos \theta \\ \rho \sin \phi \sin \theta \\ \rho \cos \phi \end{bmatrix};$$

$$\rho \geq 0, \phi \in [0, \pi], \theta \in [0, 2\pi].$$

Sturm-Liouville

$$t \in [0, L], \lambda \in \mathbb{R}, \begin{cases} u''(t) + \lambda u(t) = 0 \\ u(0) = u(L) = 0 \end{cases}$$

if $\lambda \leq 0$: $u(t) = 0$

if $\lambda > 0$: $u(t) = \sin(t\sqrt{\lambda}) \frac{u_1}{\sqrt{\lambda}}, \sqrt{\lambda} = \frac{n\pi}{L}$

Heat (finite)

$$x \in [0, L], t > 0, \begin{cases} \frac{\partial u}{\partial t} = a^2 \frac{\partial^2 u}{\partial x^2} \\ u(0, t) = u(L, t) = 0 \\ u(x, 0) = f(x) \end{cases}$$

$$u(x, t) = \sum_{n \geq 1} \alpha_n \sin\left(\frac{n\pi x}{L}\right) \exp\left(-\left(\frac{an\pi}{L}\right)^2 t\right)$$

$$\text{with } \alpha_n = \frac{2}{L} \int_0^L f(x) \sin\left(\frac{n\pi x}{L}\right) dx$$

Heat (infinite)

$$x \in \mathbb{R}, t > 0, \begin{cases} \frac{\partial u}{\partial t} = a^2 \frac{\partial^2 u}{\partial x^2} \\ u(x, 0) = f(x) \end{cases}$$

$$u(x, t) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} \hat{f}(\alpha) \exp(i\alpha x - (a\alpha)^2 t) d\alpha$$

Divergence Theorem (Plane) Let $\Omega \subseteq \mathbb{R}^2$ be a regular domain with a positively oriented boundary $\partial\Omega$, and $F \in C^1(\overline{\Omega}, \mathbb{R}^2)$. Then

$$\iint_\Omega \operatorname{div} F(x, y) dx dy = \int_{\partial\Omega} F \cdot n dl = \int_a^b \langle F(\gamma(t)), (\gamma'_2(t), -\gamma'_1(t)) \rangle dt.$$

Stokes' Theorem Let $\Omega \subseteq \mathbb{R}^3$ be an open set, $\Sigma \subseteq \Omega$ a piecewise smooth orientable surface, and $F \in C^1(\Omega, \mathbb{R}^3)$, then

$$\iint_\Sigma \operatorname{curl} F \cdot n ds = \int_{\partial\Sigma} F \cdot d\mathbf{l} = \iint_A \left\langle \operatorname{curl} F(\sigma(u, v)), \frac{\partial \sigma}{\partial u} \wedge \frac{\partial \sigma}{\partial v} \right\rangle du dv$$

Let $\Omega \subseteq \mathbb{R}^n$.

Convex if $\forall x, y \in \Omega$, the line segment from x to y is within Ω .

Star-shaped if $\exists z \in \Omega$ such that $\forall x \in \Omega$, the line segment from z to x is within Ω . $\operatorname{rot}(\operatorname{rot} \vec{F}) = -\Delta \vec{F} + \operatorname{grad}(\operatorname{div} \vec{F})$

$$\operatorname{div}(f \operatorname{grad} g) = f \Delta g + \nabla f \cdot \nabla g$$

$$\nabla \cdot (f \vec{A}) = (\nabla f) \cdot \vec{A} + f(\nabla \cdot \vec{A})$$

$$\Delta(fg) = f \Delta g + 2 \nabla f \cdot \nabla g + g \Delta f$$

3. Wave (finite):

$$x \in [0, L], t > 0, \begin{cases} \frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2} \\ u(0, t) = u(L, t) = 0 \\ u(x, 0) = f(x) \\ \frac{\partial u}{\partial t}(x, 0) = g(x) \end{cases}$$

$$u(x, t) = \sum_{n \geq 1} \left[a_n \cos\left(\frac{n\pi ct}{L}\right) + b_n \sin\left(\frac{n\pi ct}{L}\right) \right] \sin\left(\frac{n\pi x}{L}\right)$$

with

$$a_n = \frac{2}{L} \int_0^L f(x) \sin\left(\frac{n\pi x}{L}\right) dx,$$

$$b_n = \frac{2}{n\pi c} \int_0^L g(x) \sin\left(\frac{n\pi x}{L}\right) dx$$

Piecewise Continuity & Differentiability

$f : [a, b] \rightarrow \mathbb{R}$ is *piecewise continuous* if there is a partition

$$a = a_0 < a_1 < \cdots < a_n = b$$

such that $\lim_{x \rightarrow a_i^-} f(x)$ and $\lim_{x \rightarrow a_i^+} f(x)$ exist (finite).

Similarly, f is *piecewise C^1* if it is continuously differentiable on each open subinterval and the one-sided derivatives at boundaries exist.

Euler’s Formulas

$$e^{x+iy} = e^x (\cos y + i \sin y), \quad \sin x = \frac{e^{ix} - e^{-ix}}{2i}, \quad \cos x = \frac{e^{ix} + e^{-ix}}{2}.$$

Orthogonality (Sine/Cosine Products)

For $n, m \in \mathbb{N}_{\geq 1}$ and period $T > 0$:

$$\frac{2}{T} \int_0^T \cos\left(\frac{2\pi n}{T}x\right) \cos\left(\frac{2\pi m}{T}x\right) dx = \begin{cases} 1 & n = m, \\ 0 & n \neq m \end{cases}$$

(Same for $\sin \sin$, and $\cos \sin$ integrates to 0.)

Integration Over One Period

If f is T -periodic and piecewise continuous, then for any $a \in \mathbb{R}$:

$$\int_a^{a+T} f(x) dx = \int_0^T f(x) dx.$$

Dirichlet’s Theorem (Pointwise Convergence)

Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be T -periodic and piecewise C^1 . Then, for all $x \in \mathbb{R}$,

$$Ff(x) = \lim_{t \rightarrow 0} \frac{f(x-t) + f(x+t)}{2}.$$

Real Fourier Series

For $f : \mathbb{R} \rightarrow \mathbb{R}$, T -periodic, piecewise C^1 , the real Fourier series is

$$Ff(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} \left[a_n \cos\left(\frac{2\pi n}{T}x\right) + b_n \sin\left(\frac{2\pi n}{T}x\right) \right].$$

Fourier Coefficients:

$$a_n = \frac{2}{T} \int_0^T f(x) \cos\left(\frac{2\pi n}{T}x\right) dx, \quad b_n = \frac{2}{T} \int_0^T f(x) \sin\left(\frac{2\pi n}{T}x\right) dx, \\ a_0 = \frac{2}{T} \int_0^T f(x) dx.$$

Parity: If f is even, $b_n = 0$; if f is odd, $a_n = 0$.

Term-by-Term Differentiation

If f is T -periodic, continuous, and piecewise C^1 , then

$$\frac{d}{dx} [Ff(x)] = \sum_{n=1}^{\infty} \frac{2\pi n}{T} \left[-a_n \sin\left(\frac{2\pi n}{T}x\right) + b_n \cos\left(\frac{2\pi n}{T}x\right) \right] \\ = \lim_{t \rightarrow 0} \frac{f'(x-t) + f'(x+t)}{2}.$$

Term-by-Term Integration

If f is T -periodic, continuous, and piecewise C^1 , then

$$\int Ff(x) dx = \sum_{n=1}^{\infty} \frac{T}{2n\pi} \left[a_n \sin\left(\frac{2\pi n}{T}x\right) - b_n \cos\left(\frac{2\pi n}{T}x\right) \right] + C \\ = \lim_{h \rightarrow 0} \frac{1}{2h} \int_{x-h}^{x+h} Ff(t) dt,$$

where C is the constant of integration.

4. Laplace (rectangle)

$$x \in [0, L], y \in [0, M], \quad \begin{cases} \Delta u = \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0 \\ u(x, 0) = \alpha(x), \quad u(x, M) = \beta(x) \\ u(0, y) = \gamma(y), \quad u(L, y) = \delta(y) \end{cases}$$

$$\mathbf{u}(\mathbf{x}, \mathbf{y}) = v(x, y) + w(x, y) \text{ with} \\ \mathbf{w}(\mathbf{x}, \mathbf{y}) = \sum_{n \geq 1} \left(a_n \cosh\left(\frac{n\pi y}{L}\right) + b_n \sinh\left(\frac{n\pi y}{L}\right) \right) \sin\left(\frac{n\pi x}{L}\right) \\ \text{with} \\ - a_n = \frac{2}{L} \int_0^L \alpha(x) \sin\left(\frac{n\pi x}{L}\right) dx, \\ - b_n \sinh\left(\frac{n\pi M}{L}\right) = \frac{2}{L} \int_0^L \beta(x) \sin\left(\frac{n\pi x}{L}\right) dx - a_n \cosh\left(\frac{n\pi M}{L}\right) \\ \mathbf{v}(\mathbf{x}, \mathbf{y}) = \sum_{n=1}^{\infty} \left[d_n \cosh\left(\frac{n\pi}{M}x\right) + e_n \sinh\left(\frac{n\pi}{M}x\right) \right] \sin\left(\frac{n\pi}{M}y\right) \\ \text{with} \\ - d_n = \frac{2}{M} \int_0^M \gamma(y) \sin\left(\frac{n\pi}{M}y\right) dy \\ - e_n \sinh\left(\frac{n\pi}{M}L\right) = \frac{2}{M} \int_0^M \delta(y) \sin\left(\frac{n\pi}{M}y\right) dy - d_n \cosh\left(\frac{n\pi}{M}L\right)$$

$$y'(x) + p(x)y(x) = f(x) \Rightarrow y(x) = ce^{-P(x)} + \left(\int f(x)e^{P(x)} dx \right) e^{-P(x)}, \\ P(x) = \int p(x) dx$$

$$y''(x) + py'(x) + qy(x) = 0, \lambda^2 + p\lambda + q = 0 \\ - \Delta > 0, \lambda_1 \neq \lambda_2 \Rightarrow y_{\text{hom}}(x) = c_1 e^{\lambda_1 x} + c_2 e^{\lambda_2 x} \\ - \Delta = 0, \lambda_1 = \lambda_2 = \lambda \Rightarrow y_{\text{hom}}(x) = c_1 e^{\lambda x} + c_2 x e^{\lambda x} \\ - \Delta < 0, \lambda = \alpha \pm i\beta \Rightarrow y_{\text{hom}}(x) = c_1 e^{\alpha x} \cos(\beta x) + c_2 e^{\alpha x} \sin(\beta x)$$

Complex Fourier Coefficient

Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be T -periodic and piecewise continuous. The complex Fourier coefficients are:

$$c_n = \frac{1}{T} \int_0^T f(x) e^{-i\frac{2\pi}{T}nx} dx, \quad Ff(x) = \sum_{n=-\infty}^{\infty} c_n e^{i\frac{2\pi n}{T}x}.$$

For $\phi : \mathbb{R} \rightarrow \mathbb{C}$,

$$\int_a^b \phi(x) dx = \int_a^b \text{Re}(\phi(x)) dx + i \int_a^b \text{Im}(\phi(x)) dx.$$

Relation to (a_n, b_n)

$$c_n = \frac{1}{2}(a_n - i b_n), \quad c_{-n} = \frac{1}{2}(a_n + i b_n), \quad c_0 = \frac{a_0}{2}. \\ a_n = c_n + c_{-n} \quad a_0 = 2c_0 \quad b_n = i(c_n - c_{-n})$$

Fourier Series on $[0, L]$

For $f : [0, L] \rightarrow \mathbb{R}$ (piecewise C^1):

$$F_c f(x) = \frac{\tilde{a}_0}{2} + \sum_{n=1}^{\infty} \tilde{a}_n \cos\left(\frac{\pi n}{L}x\right), \quad \tilde{a}_n = \frac{2}{L} \int_0^L f(x) \cos\left(\frac{\pi n}{L}x\right) dx.$$

$$F_s f(x) = \sum_{n=1}^{\infty} \tilde{b}_n \sin\left(\frac{\pi n}{L}x\right), \quad \tilde{b}_n = \frac{2}{L} \int_0^L f(x) \sin\left(\frac{\pi n}{L}x\right) dx.$$

Parseval’s Identity (Periodic Case)

If f is T -periodic (piecewise C^1),

$$\frac{2}{T} \int_0^T f^2(x) dx = \frac{a_0^2}{2} + \sum_{n=1}^{\infty} (a_n^2 + b_n^2) = 2 \sum_{n=-\infty}^{\infty} |c_n|^2.$$

Plancherel Theorem

Let $f \in L^2(\mathbb{R})$. Then its Fourier transform $\hat{f}$ is also in $L^2(\mathbb{R})$, and:

$$\int_{-\infty}^{\infty} |f(x)|^2 dx = \int_{-\infty}^{\infty} |\hat{f}(\xi)|^2 d\xi.$$

The Fourier Transform

If $f : \mathbb{R} \rightarrow \mathbb{R}$ with $\int_{-\infty}^{\infty} |f(x)| dx < \infty$, its (unitary) Fourier transform is

$$\hat{f}(\alpha) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(x) e^{-i\alpha x} dx,$$

Inverse Transform

If $\varphi(\alpha)$ is similarly integrable,

$$\mathcal{F}^{-1}(\varphi)(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \varphi(\alpha) e^{i\alpha x} d\alpha.$$

Convolution Product

Let $f, g : \mathbb{R} \rightarrow \mathbb{R}$ such that $\int_{-\infty}^{+\infty} |f(x)| dx < +\infty, \int_{-\infty}^{+\infty} |g(x)| dx < +\infty$.

$$(f * g)(x) = \int_{-\infty}^{+\infty} f(x-t) g(t) dt = \int_{-\infty}^{+\infty} f(t) g(x-t) dt$$

Scaling

$$\mathcal{F}\{f(ax)\} = \frac{1}{|a|} \hat{f}\left(\frac{\alpha}{a}\right)$$

Differentiation

$$\mathcal{F}\left\{\frac{d^n}{dx^n} f(x)\right\} = (i\alpha)^n \hat{f}(\alpha)$$

Shifting

$$\mathcal{F}\{f(x - x_0)\} = e^{-i\alpha x_0} \hat{f}(\alpha)$$

Integration

$$\mathcal{F}\left\{\int_{-\infty}^x f(\xi) d\xi\right\} = \frac{1}{i\alpha} \hat{f}(\alpha), \quad \alpha \neq 0$$

Modulation

$$\mathcal{F}\{e^{i\omega_0 x} f(x)\} = \hat{f}(\alpha - \omega_0)$$

Differentiation of Transform

$$\mathcal{F}((-ix)^n f(x))(\alpha) = \frac{\partial^n}{\partial \alpha^n} \hat{f}(\alpha)$$

Convolution

$$\mathcal{F}[f * g](\alpha) = \sqrt{2\pi} \hat{f}(\alpha) \hat{g}(\alpha)$$

Multiplication

$$\mathcal{F}\{f(x) \cdot g(x)\} = \frac{1}{\sqrt{2\pi}} (\mathcal{F}\{f(x)\} * \mathcal{F}\{g(x)\})$$

Important Trigonometric Identities

$$\sin(2x) = 2 \sin x \cos x, \\ \cos(2x) = 2 \cos^2 x - 1 = 1 - 2 \sin^2 x = \cos^2 x - \sin^2 x, \\ \cos(a \pm b) = \cos a \cos b \mp \sin a \sin b, \\ \sin(a \pm b) = \sin a \cos b \pm \cos a \sin b, \\ \cos a \cos b = \frac{1}{2} [\cos(a - b) + \cos(a + b)], \\ \sin a \sin b = \frac{1}{2} [\cos(a - b) - \cos(a + b)], \\ \sin a \cos b = \frac{1}{2} [\sin(a + b) + \sin(a - b)], \\ \cos a \sin b = \frac{1}{2} [\sin(a + b) - \sin(a - b)]. \\ \cos(n\pi) = (-1)^n$$